

HMK: Head-Mounted-Keyboard for Text Input in Virtual or Augmented Reality

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Figure 1: A) Head-mounted-keyboard while in use. B) Closer look at the custom keytops. C) Average WPM over three days

ABSTRACT

Text input is essential in a variety of uses in virtual and augmented reality (VR and AR). We present HMK: an effective text input method that mounts split keyboards on the left and right side of the head mounted display (HMD). Users who can touch-type are able to type using HMK by relying on their familiarity with the normal QWERTY keyboard. We develop custom keycaps to make it easier to find the home position. A study with three participants shows that users retain most of their normal keyboard typing skills. The participants achieved, on average, 34.7 words per minute (WPM) by the end of three days of use, retaining 81 percent of their regular entry speed.

CCS CONCEPTS

• Human-centered computing → Keyboards; Virtual reality.

KEYWORDS

Text input, VR/AR, HMD, keyboard

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1 INTRODUCTION

With the development of VR / AR technologies for use in office and communication activities, the need for an easy-to-use, efficient text input method for VR and AR devices is increasing. Most VR/AR text input methods set a pointer on a virtual keyboard, which, while it can be used by anyone, is frustrating for expert typists due to its inefficiency. Many attempts have been conducted to improve VR/AR text input, but no feasible approaches have been published to date. While users can use normal keyboards for efficient VR/AR text input, they can only be used while sitting, which constrains the interactions possible. Our proposal, Head-Mounted-Keyboard, "HMK" is a method of VR/AR text input that sets a split-keyboard on the HMD; it is both efficient, and easy to use anywhere.

2 RELATED WORK

A variety of VR/AR input methods have been developed, each with their own advantages and disadvantages. Tafadzwa and Ahmed classified and analyzed various text entry techniques for VR [1]. Physical techniques include 1) physical QWERTY, 2) mobile keypads; virtual keyboard including 1) head pointing, 2) finger, wrist, and hand gestures, 3) controller, 4) touch-based, 5) eye gaze, and 6) word-level techniques. While novel virtual techniques are used for text input in VR.

Prior studies have shown that physical keyboard methods achieve high average speed (27.4 - 44.4 WPM depending on the method [1]), but most require the user to be sitting even though VR and AR is primarily used while standing and moving around. Virtual keyboards can be used while standing and are currently the most widely used method, but generally suffer from low average speeds. For example, Marco et al. reported input speeds between 5.3 WPM to 15.4 WPM for 6 VR input methods using virtual keyboards [5]. Other novel methods, such as hand-gesture methods have low average speed and require users to learn entirely new, often non-intuitive text input styles. Some methods of using hand tracking such as PinchType

[2] can be fast, but they rely heavily on having auto-correct and require much training before use.

Other than input methods designed specifically for VR and AR use, general wearable input methods have also been widely researched and some can be used for VR and AR. Torso Keyboard is a wearable text entry method that places split keyboards onto a vest or jacket [4]. Subjects were able to input over 30 WPM on various typing tasks while walking and standing. While it is indeed similar in concept, HMK is directly attached to the HMD being worn, enabling user to quickly find the keyboard position due to the fact that the keyboard location never changes. In contrast, Torso Keyboard attaches the keyboard to a vest that moves, forcing users to interrupt what they are doing to look at the keyboard or feel for its location every time they want to use it, which could degrade the VR/AR experience.

3 PROPOSED METHOD

Head-mounted-keyboard (HMK) is an input method designed to be used in mobile VR or AR environments. A pair of split keyboards is affixed on the left and right sides of an HMD. The HMD user raises their hands to press the keyboard to input text, see Fig. 1A. Even though the user cannot see the keyboard, they can start using it quickly as the keyboard coincides with the sides of their head. HMK uses a modified QWERTY layout keyboard which most people are familiar with, and high performance is anticipated. HMK is effective in both VR and AR environments. Since it does not require a controller to be held, users can directly interact with physical objects. This means HMK is especially effective in AR environments.

HMK best supports the expert user who is proficient in touch-typing with the QWERTY layout. We assume that HMK is not for continuous typing sessions such as writing a report but intermittent and short texting such as annotating surrounding objects or chatting. Although it appears to be effective, there is still a concern. Can the user quickly find the key home position on the keyboard without needing to see it? In order to address this concern, we replaced the keycaps with custom ones that assist in finding the home position.

3.1 Implementation

Our prototype uses the lightweight Corne Cherry split QWERTY keyboard, which is commercially available; each half is attached on either side of an Oculus Quest 2, and connected by a USB cable. We selected this unit because of its light weight (285g). The Corne keyboard has a column staggered layout unlike the usual row staggered layout, meaning keys are offset vertically instead of horizontally. This might confuse users and some training is needed.

We used custom keycaps to make the keys easy to differentiate, see Fig. 1B. Home position (ASDF and JKL; keys) have flat surfaces, with the keys surrounding it curved inward making the keyboard surface into a short bowl-like shape. Furthermore, F and J keys, on which the index finger rests, are made deeper than the others with a small budge in the middle for even easier identification. With these haptic clues, even new users can quickly locate the home position.

4 USER STUDY

To test the performance of our method, we performed a user study that compared 3 input methods in a VR environment while wearing an HMD; 1) normal keyboard on the desk, 2) HMK, and 3) controller pointing using raycasting. The user study had three participants with each participant joining for three days. The participants had sufficient experience and skill to be able to type without looking at their hands. They did the experiment while sitting and wearing the Oculus Quest 2. They had to type without looking at either their hands or keyboard. 10 minutes were used to familiarize themselves with the input methods before starting the experiment. Participants were then asked to type using the normal keyboard while wearing HMD to determine a baseline, next with HMK, and lastly with controller pointing on standard virtual keyboard from Oculus as a comparison. It took a total approximately 20 minutes per day for each person. During the experiment, participants were asked to type a set of random English phrases [3] within 3 minutes, and to correct all errors before continuing on to the next phrase. Errors were shown in red in the user input area for ease of identification. Total keystrokes and backspace were then used to calculate typing speed (WPM) and error rate, where 1 word is defined as 5 keystrokes, not including backspace, and the error rate is defined as the number of backspaces compared to total keystrokes.

4.1 Result and User feedback

The results of the experiment shown in Fig. 1C are as follows. Participants were able to type up to 70 to 81 percent of their normal keyboard typing speed with HMK, resulting in average of 26.7 WPM during the first session and increasing to 34.7 WPM by the third session. The controller pointing method averaged around 17.4 WPM during the first session and 17.0 WPM in the third session, for 40 to 46 percent of their normal keyboard typing speed. Participants also improved their typing speed with HMK over the 3 sessions, whereas no improvement was seen with controller pointing method.

The average error rate of the normal keyboard was 7.6 percent, HMK 11.7 percent, and controller pointing 7.4 percent. Data was insufficient to discern any pattern across the three sessions, but HMK consistently had the highest error rate.

Based on user feedback after testing, participants noted a few problems such as: 1) Having to raise their hands to type is tiring when used for a long time. 2) M and N, V and B, or T and R were frequent mistypes because of the different key position due to the unusual column staggered keyboard layout.

With regard to the first point, we do not intend for long-term use. The second problem can be solved naturally with practice. However, we can also reduce the problem by redesigning the keyboard layout and shape exclusively for HMK, revising the custom keycaps design or even the keyboard base.

5 CONCLUSION

We proposed HMK, a text entry method for VR and AR environments. A prototype was developed using a commercially available split keyboard and customizing its keycaps. Based on a user study with the prototype, we conclude that it has high entry rates. Improvements in accuracy and usability are possible with further design enhancements.

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